Soknarith (Soki) Sem

(401) 952–6819 — seminaia401@gmail.com — Seminaia — soknarith-sem — Worcester, MA

Summary

Systems modeling and simulation engineer with a strong foundation in MATLAB-based modeling, numerical methods, and physics-based simulation. Experience developing computational models across multiple scales, from atomistic simulations to dynamic system analysis. Skilled in translating real-world physical behavior into computational frameworks for analysis, validation, and optimization.

Work Experience

Research Assistant / Graduate Teaching Assistant — **May 2024 to Present**
Worcester Polytechnic Institute – Worcester, MA

- Developed physics-based computational models using DFT and molecular dynamics to analyze system behavior and transport mechanisms.
- Applied numerical methods and data analysis to evaluate system response across varying operating conditions.
- Utilized MATLAB and Python for simulation post-processing, data analysis, and visualization of complex system behavior.
- Translated physical phenomena into computational frameworks to support predictive modeling and system-level understanding.

Chemical Process Engineer — **May 2023 to May 2024**
Knowles Precision Devices – New Bedford, MA

- Supported the R&D-to-production transition of high-voltage capacitor dielectric fluids, contributing to process scale-up and manufacturing readiness.
- Optimized manufacturing layout and process flow using Lean principles and statistical process control, reducing production costs by 15%.
- Authored SOPs and work instructions for material handling and process scale-up to support technician training and manufacturing consistency.
- Investigated material- and process-related failures through root cause analysis, deviation support, and CAPA implementation to improve process reliability.
- Modeled and analyzed process behavior using data-driven and statistical approaches to improve system performance and efficiency.

R&D Technician — **Mar 2023 to May 2023**
Factorial Energy – Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments to support experimental evaluation under varied operating conditions.
- Reduced material waste by 20% through redesign of the inventory system.

Downstream Purification Engineer — **Jul 2022 to Mar 2023**
Eurofins Scientific – Framingham, MA

- Executed purification workflows for therapeutic antibodies and enzymes using AKTA Avant systems with Unicorn software, achieving 95–99% purity.
- Applied DOE methods to optimize buffer conditions and flow rate for improved yield and purity at lab scale.
- Performed membrane filtration and chromatography studies to support pilot-scale development and manufacturing scale-up.
- Collected and interpreted analytical data using SDS-PAGE, HPLC/MS, GC/MS, and UV-Vis to confirm purity, identity, and process performance.

Education

M.S. Materials Science & Engineering — **GPA: 3.50**
Worcester Polytechnic Institute – Worcester, MA — **2024 to 2026**

B.S. Chemical Engineering; B.S. Applied Mathematics — **GPA: 3.16**
University of Rhode Island – Kingston, RI — **2019 to 2022**

Minors: Chemistry, Physics
Relevant Coursework: Advanced Process Engineering (MATLAB/Simulink simulation)

Technical Skills

Modeling & Simulation:	MATLAB, Simulink, dynamic systems modeling, numerical methods, differential equations, process modeling
Computational Methods:	DFT (VASP, GPAW), Molecular Dynamics (LAMMPS), simulation post-processing, data analysis, visualization
Software & Tools:	Python, ASPEN Plus Dynamics, Linux
Engineering Analysis:	DOE, SPC, system optimization, process optimization, technical documentation